

PROJECT DEVELOPMENT PHASE

Data Preprocessing Steps, Business Questions and Visualization

Date	01 JULY 2026
Team ID	
Project Name	Strategic Product Placement Analysis
Maximum Marks	4 Marks

Introduction

The **Project Development Phase** is the implementation stage of the Strategic Product Placement Analysis project. During this phase, the retail sales dataset was collected, cleaned, analyzed, and transformed into interactive visualizations using Tableau Desktop. The project focused on understanding product placement strategies, customer purchasing behavior, promotional effectiveness, and sales performance.

The developed dashboard provides meaningful business insights that assist retail managers in making informed decisions regarding product positioning and sales optimization.

Development Methodology

The project was developed using a structured approach consisting of multiple stages, from dataset preparation to dashboard publishing.

Development Workflow

Step	Activity	Output
Step 1	Collect Product_Positioning.csv dataset	Retail sales dataset
Step 2	Clean and validate the dataset	Error-free dataset
Step 3	Import dataset into Tableau Desktop	Connected data source
Step 4	Create calculated fields and KPIs	Business metrics
Step 5	Develop charts and visualizations	Interactive charts
Step 6	Design dashboard	Retail analytics dashboard
Step 7	Create Tableau Story	Sequential business insights

Step	Activity	Output
Step 8	Publish dashboard on Tableau Public	Online dashboard
Step 9	Upload project files to GitHub	Project repository

Dataset Details

Attribute	Description
Dataset Name	Product_Positioning.csv
File Format	CSV
Number of Records	1000
Number of Attributes	9
Domain	Retail Sales Analytics

Dataset Attributes

Field Name	Description
Product ID	Unique product identifier
Product Position	Shelf position of the product
Price	Selling price
Competitor Price	Competitor product price
Promotion	Promotional offer status
Foot Traffic	Customer traffic level
Consumer Demographics	Customer category
Product Category	Category of product
Seasonal	Seasonal availability
Sales Volume	Number of products sold

Dashboard Development

The dashboard was designed to provide an interactive overview of retail sales performance. Different visualization techniques were used to analyze product placement and customer purchasing behavior.

Dashboard Components

Visualization	Purpose
KPI Cards	Display Total Sales, Average Sales, Categories, and Products
Bar Chart	Product Category vs Sales Volume
Scatter Plot	Competitor Price vs Product Price
Heat Map	Customer Demographics vs Product Category
Bar Chart	Promotion Impact on Sales
Scatter Plot	Foot Traffic vs Sales Volume
Treemap	Product Category Distribution
Filters	Interactive data filtering

Tableau Story Development

A Tableau Story was created to present analytical findings in a logical sequence. Each story point highlights a specific business insight and guides users through the analysis process.

Story Point	Description
Story 1	Overview of Retail Sales
Story 2	Product Category Performance
Story 3	Customer Demographic Analysis
Story 4	Promotion Impact
Story 5	Competitor Price Analysis

Story Point	Description
Story 6	Seasonal Sales Trends
Story 7	Business Recommendations

Tools and Technologies Used

Technology	Purpose
Tableau Desktop	Dashboard development
Tableau Public	Dashboard publishing
GitHub	Project repository
Microsoft Word	Documentation
CSV Dataset	Retail sales data

Project Outputs

Output	Description
Interactive Dashboard	Retail sales analysis dashboard
Tableau Story	Business insights presentation
Tableau Public Dashboard	Online dashboard
GitHub Repository	Source files and documentation
Final Project Report	Project documentation

Development Challenges

Challenge	Solution
Data cleaning	Validated and formatted dataset before analysis
Dashboard design	Selected suitable charts and KPIs
Story creation	Organized insights into a logical sequence

Challenge	Solution
Tableau Public publishing	Successfully published the final dashboard after resolving publishing issues
Documentation	Prepared reports according to the project templates

Expected Outcomes

- Interactive visualization of retail sales data.
 - Improved understanding of customer purchasing behavior.
 - Better product placement recommendations.
 - Data-driven business decision-making.
 - Easy sharing through Tableau Public.
 - Proper documentation and version control using GitHub.
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Conclusion

The Project Development Phase successfully transformed the retail sales dataset into an interactive Tableau dashboard and story. The developed solution enables retail managers to analyze product performance, customer behavior, promotional effectiveness, and seasonal trends through visual analytics. The completed project was documented, published on Tableau Public, and maintained in a GitHub repository, providing a complete and practical business intelligence solution for retail sales analysis.